$\rho^\star$

$N^\star$

$D^\star$

Reproducing Kaplan et al.

$r = -0.74$ $(-0.81, -0.65)$
$\rho^\star(C_C) = 0.005$ (0.002, 0.02)

$a = 0.864$ $(0.82, 0.90)$
$N^\star(C_C) = 4\text{T}$ (2T, 7T)

$b = 0.138$ $(0.11, 0.18)$
$D^\star(C_C) = 23\text{B}$ (14B, 45B)

Counting last layer FLOPs

$r = -0.4$ $(-0.45, -0.32)$
$\rho^\star(C_C) = 0.19$ (0.09, 0.62)

$a = 0.699$ $(0.66, 0.72)$
$N^\star(C_C) = 695\text{B}$ (412B, 1T)

$b = 0.301$ $(0.28, 0.34)$
$D^\star(C_C) = 139\text{B}$ (98B, 240B)

Correcting warmup

$r = -0.203$ $(-0.26, -0.12)$
$\rho^\star(C_C) = 1.25$ (0.58, 4.37)

$a = 0.603$ $(0.57, 0.63)$
$N^\star(C_C) = 287\text{B}$ (171B, 435B)

$b = 0.398$ $(0.37, 0.43)$
$D^\star(C_C) = 350\text{B}$ (220B, 567B)

Cosine decay

$r = -0.138$ $(-0.21, -0.06)$
$\rho^\star(C_C) = 3.05$ (0.93, 10.66)

$a = 0.574$ $(0.54, 0.61)$
$N^\star(C_C) = 195\text{B}$ (116B, 326B)

$b = 0.426$ $(0.40, 0.46)$
$D^\star(C_C) = 505\text{B}$ (309B, 823B)

Optimizer tuning (no decay)

$r = -0.0368$ $(-0.08, 0.02)$
$\rho^\star(C_C) = 10.03$ (4.94, 21.39)

$a = 0.518$ $(0.49, 0.54)$
$N^\star(C_C) = 99\text{B}$ (69B, 141B)

$b = 0.483$ $(0.46, 0.51)$
$D^\star(C_C) = 1\text{T}$ (725B, 1T)

Compute C [FLOPs]

95% confidence region
Power law fit
Hoffmann law
Kaplan law
Observations
Warmup period